him, has already happened. Again, Chewie won't know the outcome of the election for billions of years, since that's how long it will take our television signals to reach him. But when word of the election results reaches Chewie's descendants and they use it to update Chewie's flip-card book of history, his collection of past now-lists, they will find that the election results belong on the same now-list in which Chewie got up and started walking toward earth—a now-list, Chewie's descendants note, that occurs just a moment after one that contains you, in the early years of earth's twenty-first century, finishing this paragraph.

This example highlights two important points. First, although we are used to the idea that relativistic effects become apparent at speeds near that of light, even at low velocities relativistic effects can be greatly amplified when considered over large distances in space. Second, the example gives insight into the issue of whether spacetime (the loaf) is really an entity or just an abstract concept, an abstract union of space right *now* together with its history and purported future.

You see, Chewie's conception of reality, his freeze-frame mental image, his conception of what exists *now,* is every bit as real for him as our conception of reality is for us. So, in assessing what constitutes reality, it would be stunningly narrow-minded if we didn't also include his perspective. For Newton, such an egalitarian approach wouldn't make the slightest difference, because, in a universe with absolute space and absolute time, everyone's now-slice coincides. But in a relativistic universe, our universe, it makes a big difference. Whereas our familiar conception of what exists right now amounts to a single now-slice—we usually view the past as gone and the future as yet to be—we must augment this image with Chewie's now-slice, a now-slice that, as the discussion revealed, can differ substantially from our own. Furthermore, since Chewie's initial location and the speed with which he moves are arbitrary, we should include the now-slices associated with all possibilities. These now-slices, as in our discussion above, would be centered on Chewie's—or some other real or hypothetical observer's—initial location in space and would be rotated at an angle that depends on the velocity chosen. (The only restriction comes from the speed limit set by light and, as explained in the notes, in the graphic depiction we are using this translates into a limit on the rotation angle of 45 degrees, either clockwise or counterclockwise.) As you can see, in Figure 5.5, the